



Lab 9: TCP/IP Addressing and Subnetting

Introduction to TCP/IP Addressing

The TCP/IP protocol suite is the backbone of modern networking, enabling devices to communicate over the internet and local networks. Addressing is a foundational concept, ensuring data reaches its intended destination. Two versions of IP addressing are in use today: IPv4 (32-bit addresses) and IPv6 (128-bit addresses). To enable efficient use of IPv4 addresses and support different network sizes, subnetting is used to divide large networks into smaller, manageable sub-networks.

IPv4 Addressing and Subnetting

An IPv4 address consists of four octets (e.g., 192.168.1.1) and is classified into five classes:

- **Class A (1.0.0.0 - 126.255.255.255)**
 - Supports very large networks.
- **Class B (128.0.0.0 - 191.255.255.255):**
 - Supports medium-sized networks.
- **Class C (192.0.0.0 - 223.255.255.255):**
 - Supports small networks.
- **Class D (224.0.0.0 - 239.255.255.255):**
 - Used for multicast.
- **Class E (240.0.0.0 - 255.255.255.255):**
 - Reserved for research.

Each IP address consists of a network portion, which identifies the network, and a host portion, which identifies the specific device in the network. Subnetting is the process of dividing a large network into smaller sub-networks to improve efficiency and security. The subnet mask determines how many bits are used for the network and how many for hosts. Subnetting benefits include efficient IP address usage, improved security, and reduced network congestion. A more detailed explanation of IP Addressing and subnetting can be found [here](#).

IPv6 Addressing and Dual-Stack Networking

IPv4 is limited to approximately 4.3 billion addresses, which is insufficient due to global internet expansion. IPv6 provides 128-bit addressing (supporting 340 undecillion addresses), simplified address assignment through auto-configuration, and better security with built-in IPSec.



encryption. IPv6 addresses are written in hexadecimal and consist of 8 groups of 16-bit values, separated by colons (:), for example 2001:0db8:85a3:0000:0000:8a2e:0370:7334, is an IPv6 address. In shortened IPv6 notation, leading zeros can be omitted, and consecutive zeros can be replaced with "::". Because the transition from the limited IPv4 address space to the vast capacity of IPv6 cannot happen overnight, the internet requires a transitional bridge. This is where a dual-stack network becomes essential. In a dual-stack configuration, networking hardware and end devices are configured to run both the IPv4 and IPv6 protocols simultaneously. Each device is assigned both an IPv4 address and an IPv6 address. This allows a single device to dynamically choose the correct protocol based on its destination: it can communicate natively with older IPv4 only infrastructure without losing connection, while also taking full advantage of modern IPv6 networks when available. By running both protocols side by side, dual-stack networking ensures complete backward compatibility and prevents devices from being cut off during the long global migration to IPv6.

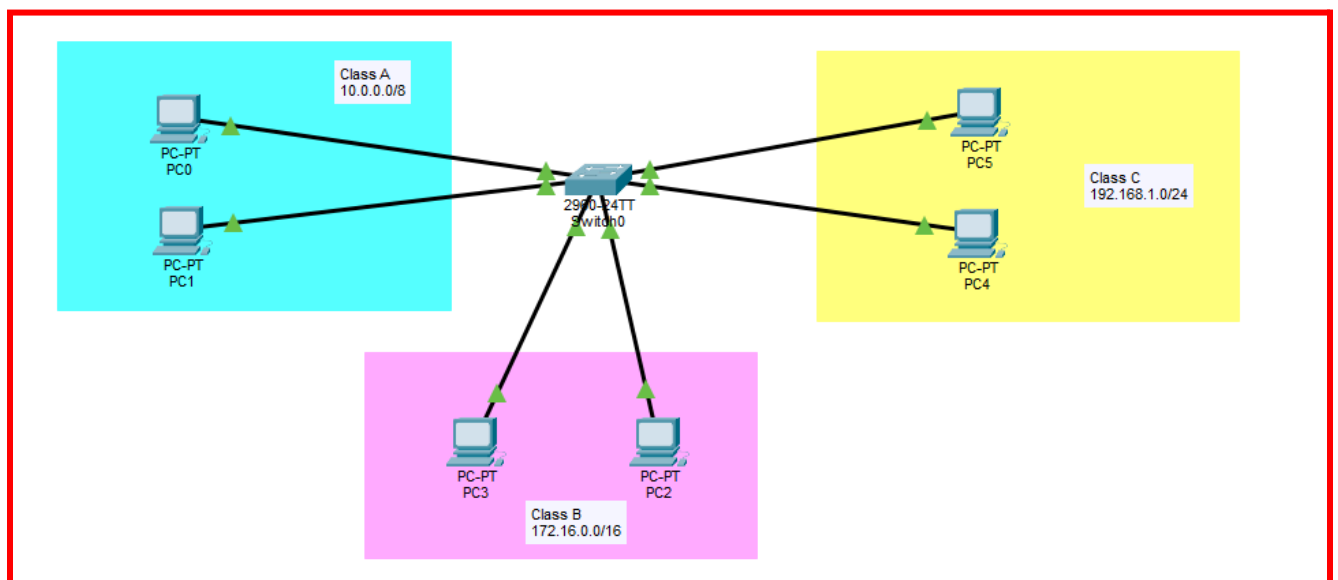
Lab Tasks

Note: All tasks must be documented in a single report. Include screenshots of every step (description is not as necessary as the screenshots).

CLO2: Apply core networking concepts and protocols, including but not limited to routing, Congestion control, and error management, to establish and analyze reliable data communication across networks.

Task 1: Classful Addressing (10 marks)

The following are the steps, topology and IP addresses you need to use for this task.





Before beginning the task, it is important to understand two special IP addresses in every subnet. The network address is the first IP address in a subnet and represents the identity of the entire network. It is used by routers to identify and route traffic to that network. The broadcast address is the last IP address in a subnet and is used to send a message to all devices within that network simultaneously. These two addresses cannot be assigned to individual devices. To understand this better, consider the Class A network used in this task. The network is 10.0.0.0/8, which means the subnet mask is 255.0.0.0. In this case, 10.0.0.0 is the network address and cannot be assigned to any PC because it represents the entire network itself. Similarly, 10.255.255.255 is the broadcast address, which is reserved for sending messages to all devices in that network. This means valid usable IP addresses for devices start from 10.0.0.1 and go up to 10.255.255.254. That is why in this task, PCs are assigned addresses like 10.0.0.1 and 10.0.0.2 instead of 10.0.0.0.

Step 1: Setting Up the Network Topology

To begin, open Cisco Packet Tracer and add one switch (2960) along with six PCs to the workspace. After placing the devices, connect each of the six PCs to the switch using copper straight-through cables.

Step 2: Assigning IPv4 Addresses

Next, configure the PCs into three separate groups representing Class A, Class B, and Class C networks. For the Class A network, assign PC1 the IP address 10.0.0.1 with a subnet mask of 255.0.0.0, and assign PC2 the IP address 10.0.0.2 with the same subnet mask. (In this case, the network address is 10.0.0.0 and the broadcast address is 10.255.255.255.) For the Class B network, assign PC3 the IP address 172.16.0.1 with a subnet mask of 255.255.0.0, and assign PC4 the IP address 172.16.0.2 with the same subnet mask. (The network address for this subnet is 172.16.0.0, while the broadcast address is 172.16.255.255.) For the Class C network, assign PC5 the IP address 192.168.1.1 with a subnet mask of 255.255.255.0, and assign PC6 the IP address 192.168.1.2 with the same subnet mask. (The network address in this case is 192.168.1.0, and the broadcast address is 192.168.1.255.)

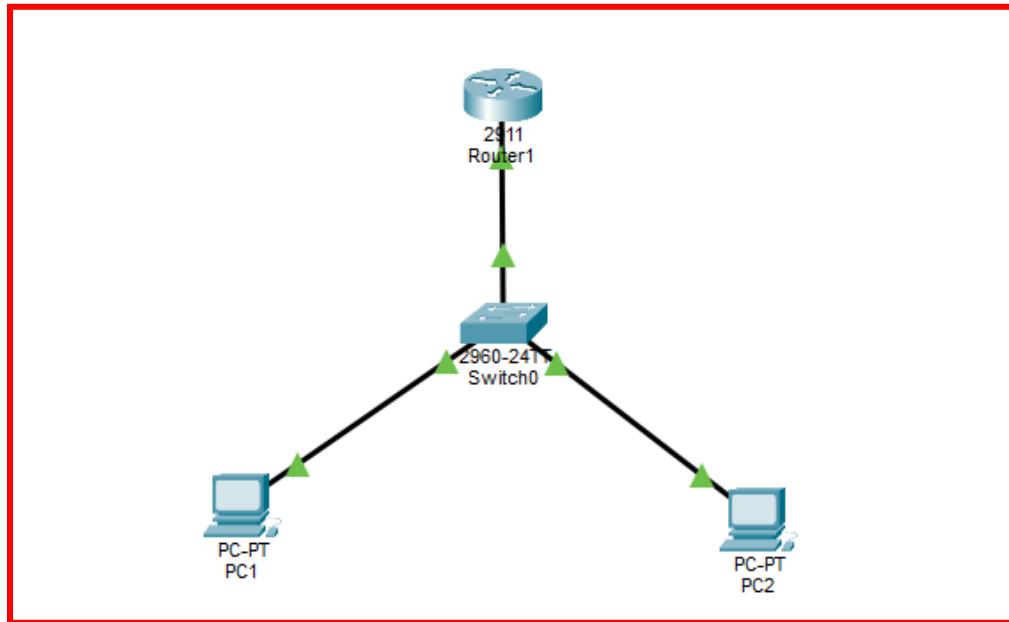
Step 3: Testing Connectivity

To test the configuration, open the command prompt on PC1 and attempt to ping PC2 by typing ping 10.0.0.2. This ping should be successful because both devices belong to the same Class A network and share the same network portion of their IP address. Next, from PC1, attempt to ping PC5 by typing ping 192.168.1.1. This ping will fail because PC1 and PC5 belong to different networks. Even though all devices are connected to the same switch, the subnet masks logically separate them into different networks, and a switch alone cannot route traffic between these networks.



Task 2: Dual Stack Network (10 marks)

The following are the steps, topology and IP addresses you need to use for this task.



Step 1: Setting Up the Network Topology

To begin, open Cisco Packet Tracer and add two PCs, one switch (2960), and one router (2911) to the workspace. Connect both PCs to the switch using copper straight-through cables. Then connect the switch to the router using another copper straight-through cable.

Step 2: Configuring IPv4 on the Router

Next, access the router's command line interface and enter global configuration mode. Configure the GigabitEthernet interface connected to the switch by assigning it an IPv4 address and enabling it.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up
Router(config-if)#exit
```



Step 2: Configuring IPv6 on the Router

After configuring IPv4, enable IPv6 routing on the router. Then assign an IPv6 address to the same interface so the router can handle both protocols.

```
Router(config)#ipv6 unicast-routing
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ipv6 address 2001:db8:1::1/64
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

Step 4: Configuring PCs with IPv4 and IPv6

Now configure both PCs with dual stack addressing. On each PC, go to the Desktop tab and open IP Configuration. For PC1, assign IPv4 address 192.168.1.2 with subnet mask 255.255.255.0 and default gateway 192.168.1.1. Then assign IPv6 address 2001:db8:1::2/64 and set the IPv6 gateway to 2001:db8:1::1. For PC2, assign IPv4 address 192.168.1.3 with subnet mask 255.255.255.0 and default gateway 192.168.1.1. Then assign IPv6 address 2001:db8:1::3/64 and set the IPv6 gateway to 2001:db8:1::1.

Step 5: Verifying Dual-Stack Connectivity

To verify IPv4 connectivity, open the command prompt on PC1 and send a ping to PC2 using its IPv4 address (ping 192.168.1.3). If configured correctly, replies should arrive. Next, verify IPv6 connectivity by sending a ping to PC2 using its IPv6 address (ping 2001:db8:1::3). This ping should also be successful, confirming that both IPv4 and IPv6 are functioning simultaneously on the network.

Submission Guidelines (5 Marks):

- Submit a single PDF document containing all tasks and the CISCO packet file(s) to the GCR, **also ensure your file name is visible in all pictures and contains your roll number.**
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **L_i231234_Lab1.**
- Name the CISCO files as follows: SectionLetter_RollNumber_TaskNumber, for example: **L_i231234_Lab1. (for multiple files add the task number at the end)**
- The unethical usage of Generative AI is strictly prohibited and will result in 0 marks.



CL3001 - Computer Networks Lab
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- In case a demo is conducted, there will be deductions based on your answers to the questions asked.

NOTE: Screenshots where your name, roll number, section AND lab number are NOT visible, will automatically be given 0/not considered valid.

Sample screenshot:

